

WatchTower: Observation-First Forensics for Multi-Agent AI Systems

Detecting Silent Failures and Report-vs-Reality Gaps When an Agent’s Self-Report Is Not Ground Truth

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Proof-of-Concept Artifact

<https://github.com/beejak/agentwatch>

Abstract—Monitoring tools for LLM-agent systems consume what an agent *reports* it did—the trace of tool calls and outputs it emits. We argue this is insufficient as a security signal: agents fail in ways their self-report does not reveal, and a compromised or buggy agent (or tool) can under-report its own actions. We characterize two structural blind spots—*silent failures* (the agent reports success while looping, burning tokens, or collapsing in output diversity) and *cross-layer discrepancies* (the agent’s reported actions disagree with an independent host view)—and present WATCHTOWER, an observation-first platform that detects them by reconciling the self-report against (a) the *shape* of the execution trajectory and (b) an independent *host* ground truth. On a frozen, held-out corpus a faithful self-report baseline detects 0% of both classes *by construction*, while WATCHTOWER detects silent failures at 0.86 recall (0.07 FP) and cross-layer discrepancies at 1.00 (0.00 FP). We confirm the mechanism on real, emergent traffic from an LLM agent whose behavior is captured through an egress proxy ($n=120$; recall 1.00, FP 0.00), and we map the detection envelope—including where detection falls off. This paper documents a proof-of-concept artifact released for reproduction and extension; WATCHTOWER observes and attributes but does not enforce (a co-designed enforcement layer is described separately).

I. INTRODUCTION

LLM agents increasingly act through tools, memory, and peer agents. Existing observability (LangSmith, Langfuse) records the agent’s emitted trace and renders it on a dashboard. That trace is the agent’s *self-report*, and a self-report is not ground truth:

- **The silent failure.** An agent retries the same call 150 times; every span is `status=ok`; cost balloons. Nothing *errored*—an output-level monitor shows a green dashboard while the task was broken throughout.
- **The lying agent.** An agent’s trace reports one network call; the host observed four. The three unreported calls are invisible to anything that trusts the trace—an exfiltration/tampering signal.

We argue monitoring must *reconcile the self-report against independent evidence*—the trajectory’s shape and an independent host view—and that the discrepancy is the signal.

Contributions.

- **C1, silent-failure detection (SC2):** cost-anomaly, retry-loop, and entropy-collapse detectors that flag failures which never raise an error.

- **C2, cross-layer reconciliation (SC3):** comparison of agent-reported network actions against host telemetry, scored as a tamper/exfil discrepancy.
- **C3, an append-only forensic Chronicle + attribution (SC1):** a no-UPDATE/no-DELETE event store enabling post-hoc attribution of a coordination failure to agent, action, and call-tree position.
- **Evidence:** a frozen, held-out evaluation with baselines and confidence intervals, an operating-envelope sweep, and confirmation on real emergent LLM-agent traffic. All artifacts are released for reproduction.

II. BACKGROUND AND RELATED WORK

Trace/output monitors (LangSmith, Langfuse) record and visualize the emitted trace; they trust the self-report and thus cannot flag an all-ok silent failure, nor detect a report-vs-reality gap (no host concept). **The observability gap** [1] argues output-level feedback is insufficient for coding agents; we operationalize that argument with concrete detectors and an evaluation. **Agentic process observability** [2] treats trajectories as process-mining targets and highlights stochasticity; it is complementary, not a security discrepancy detector. **Host EDR** (Sysmon, Falco, eBPF) observes OS-level network/process events but has no agent semantics; WATCHTOWER *bridges* host events to the agent trajectory via process GUIDs. The **MAST** taxonomy [3] informs the coordination-failure signatures used in attribution. Our novelty is *multi-layer reconciliation (self-report $\leftrightarrow$ trajectory $\leftrightarrow$ host) as a security signal*, with concrete detectors, a forensic substrate, and a held-out evaluation.

III. THREAT AND FAILURE MODEL

We consider agents that may report success while failing—non-adversarial bugs (retry storms, token burn) and adversarial masking—and a compromised/injected agent (or tool) that under-reports network activity (SC3 exfil). *In scope:* detection, attribution, forensic record. *Out of scope (companion enforcement system):* blocking, prompt-injection content classification, and cross-session taint.

IV. SYSTEM DESIGN

WATCHTOWER is a passive, append-only, 16-layer side-car (Fig. 1). Agents emit HMAC-signed Signals to a Redis stream; a Receiver verifies origin on every signal and fans

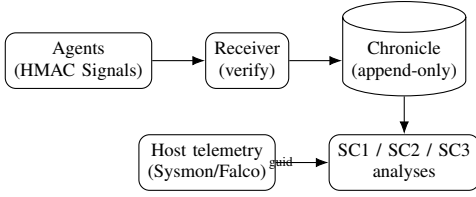


Fig. 1. Observe-and-reconcile pipeline. The Chronicle is the forensic source of truth; SC3 reconciles the agent self-report against host telemetry by process GUID.

out to inspection layers; everything is recorded to an append-only **Chronicle** (ClickHouse; no UPDATE/DELETE; 90-day TTL). Analyses run over the Chronicle.

SC1 attribution. From a trace’s spans, identify the failing agent/action and call-tree position, classified against a coordination-signature (MAST) library.

SC2 silent failure. Over a trace’s spans: a cost-anomaly ratio (actual/expected), a repeated-summary retry-loop test, and an entropy-collapse test—none of which require an error status.

SC3 cross-layer. Count agent-reported network actions; count host NetworkConnect events for the same process GUIDs; a positive delta (host > reported) is the discrepancy.

A three-stage verdict engine (deterministic → behavioral-baseline → sampled LLM judge) and a per-agent behavioral baseline (3σ ; restricted until 50 traces) support the analyses.

V. IMPLEMENTATION

Python 3.12 (all-async), Pydantic v2 (the `Signal` shape defined once), a FastAPI surface, a ClickHouse Chronicle, Redis transport, PostgreSQL baseline, Neo4j access graph, HMAC-SHA256 signing, and 12-factor environment configuration. A lightweight SDK lets any agent emit signed signals to the Chronicle or the `wt:signals` stream. Approximately 16 layers, each gated by a test. The LLM judge is a real OpenAI-compatible client when configured, and a deterministic heuristic otherwise—so the pipeline and CI run with zero LLM dependency.

VI. EVALUATION

We use frozen, human-labeled corpora; a held-out split; baselines head-to-head; and bootstrap 95% confidence intervals. Baselines: **B1** self-report (flags only an explicit error span—the faithful upper bound of any self-report-trusting monitor) and **B2** a naive cost threshold. All numbers are reproducible from the released artifacts: the synthetic corpus is `eval/corpus/traces_v0.1.jsonl` (results `eval/results/traces_v0.1_test.json`); the real-LLM corpus `eval/corpus/captured_llm_v0.1.jsonl` (`eval/results/captured_llm_test.json`); the envelope sweep `eval/results/sweep_v0.1.json`.

H3 (the spine). The self-report baseline has **0 recall** on both classes—it never sees the signal (Fig. 2). WATCHTOWER also beats naive cost (loops/entropy catch what pure cost misses; pure cost false-positives on expensive-but-correct work).

TABLE I
HELD-OUT SYNTHETIC CORPUS ($n=269$; 380-TRACE CORPUS).

Detector	SC2 recall	SC2 FPR	SC3 recall	SC3 FPR
WatchTower	0.86	0.07	1.00	0.00
Self-report (B1)	0.00	0.00	0.00	0.00
Naive cost (B2)	0.27	0.07	—	—

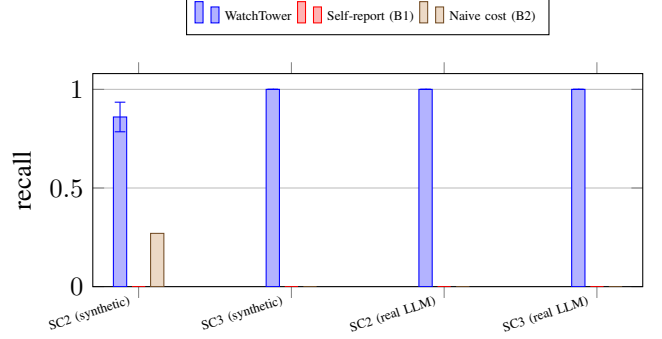


Fig. 2. Detection recall by detector, on held-out synthetic and real LLM-driven traffic. **H3**: the self-report baseline is structurally blind (0.00) across the board, while WATCHTOWER detects.

Near-real-world methodology. A DeepSeek agent (ReAct loop) acts against tasks; its tools make real HTTP egress through an independent egress proxy (mitmproxy) that records every call regardless of whether the agent emitted a span. SC2 (the model genuinely loops) and SC3 (a compromised tool’s hidden egress) *emerge* from the model’s behavior—nothing is scripted. Table III confirms H3 on real traffic. The 1.00 reflects a clean signal once the condition occurs; the conservative headline is the synthetic 0.86 of Table I, which includes borderline cases.

A. Comparison to existing monitoring

Trace/output monitors trust the self-report (no host view, no silent-failure model); host EDR sees the host but not agent intent. Only WATCHTOWER reconciles the two (Table IV).

B. Practicality

Detection is cheap and runs off the agent’s hot path (Table ??, commodity CPU, no GPU/network): all detectors are sub-millisecond per trace, and a full SC1+SC2+SC3 pass sustains $\sim 66k$ spans/s.

C. Case study: what the dashboard hides

VII. DISCUSSION AND LIMITATIONS

- **Attribution is first-error, not causal-root.** On cascade failures where the root error is not the first error span, SC1 mis-identifies the agent (Table II). Causal-root attribution is future work.
- **Synthetic realism.** The bulk corpus is generated; the robust result is the structural *baseline gap*, confirmed on real LLM-agent traffic. Absolute numbers will move on production traffic.

TABLE II
OPERATING ENVELOPE—WHERE DETECTION HOLDS AND FALLS OFF.

Condition	Boundary
SC2 silent loop	fires at loop length ≥ 10 (shorter missed)
SC2 token burn	fires at total cost $\geq \$0.12$ (FP above on costly-OK)
SC3 cross-layer	fires at delta ≥ 1 ; negative delta never fires
SC1 attribution	0.50: cascade exposes first-error $\neq$ causal-root

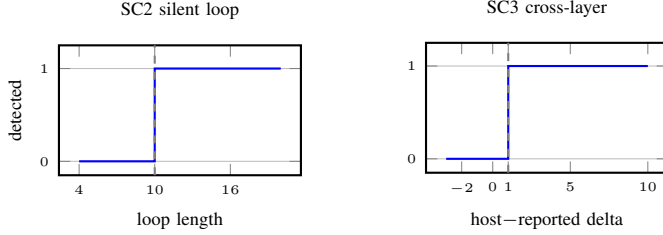


Fig. 3. Operating envelopes. SC2 silent-loop detection turns on at loop length ≥ 10 (shorter loops are honest misses); SC3 fires at host-reported delta ≥ 1 and never on negative delta (telemetry sampling). Token-burn fires at cost $\geq \$0.12$.

- **SC3 without kernel telemetry** validates HTTP egress only (proxy); raw-socket, DNS, and proxy-bypass require an eBPF/Sysmon collector (a bridge is implemented and unit-tested).
- **Captured scale.** The real-agent corpus is 120 traces—ample to confirm the mechanism, modest for headline claims; the harness scales with a parameter.
- **LLM judge is fallible**—it is the last, sampled verdict stage by design; deterministic stages are authoritative.

VIII. CONCLUSION

Treating an agent’s self-report as untrusted—and reconciling it against the trajectory and an independent host view—catches silent failures and report-vs-reality gaps that self-report-based monitoring structurally cannot. We release the implementation, frozen corpora, and harness as a proof-of-concept artifact for reproduction and extension.

ARTIFACT AVAILABILITY

All code, frozen corpora, results, and this paper are released at <https://github.com/beejak/agentwatch>. Reproduce: make eval (held-out synthetic, Table I), python -m eval.sweep (operating envelope, Table II and Fig. 3), and make capture-tier1-llm (real LLM-driven capture, Table III). Data: eval/corpus/traces_v0.1.jsonl ($n=380$) and eval/corpus/captured_llm_v0.1.jsonl ($n=120$); per-run metrics in eval/results/*.json; end-to-end demo output in examples/sample_output.json; methodology in docs/TESTING.md and docs/REAL_TRAFFIC_VALIDATION.md. The companion enforcement system is at <https://github.com/beejak/agentwatch-firewall>.

TABLE III
NEAR-REAL-WORLD: REAL LLM AGENT, EMERGENT TRAFFIC ($n=120$, 40/CLASS).

Detector	SC2 recall	SC2 FPR	SC3 recall	SC3 FPR
WatchTower	1.00	0.00	1.00	0.00
Self-report (B1)	0.00	0.00	0.00	0.00

TABLE IV
CAPABILITY COMPARISON (✓ DETECTS/PROVIDES; × CANNOT).

Capability	WatchTower	LangSmith/Langfuse	Host EDR
SC1 attribution (call-tree)	✓	partial	×
SC2 silent failure	✓	×	×
SC3 cross-layer discrepancy	✓	×	× [†]
Append-only forensic audit	✓	×	partial
Agent semantics	✓	✓	×

[†]Host EDR sees host events but has no agent self-report to reconcile against.

REFERENCES

- [1] “The Observability Gap in LLM Coding Agents,” arXiv:2603.26942, 2026.
- [2] “Agentic AI Process Observability,” arXiv:2505.20127, 2025.
- [3] Cohen et al., “MAST: A Multi-Agent System Taxonomy for LLM Failure Mode Classification,” NeurIPS, 2025.
- [4] OWASP, “Top 10 for LLM Applications,” 2025.
- [5] Microsoft, “Sysmon,” Sysinternals; Falco, “Cloud-native runtime security.”

